

Eenheid 3.5

Note Title

2016/04/14

Implisiete differentiatie

Stewart p 208

VOORBEELD: Als $y = (1 + \cos x)^{10}$

dan is $\frac{d}{dx} (1 + \cos x)^{10} = 10 \underbrace{(1 + \cos x)^9}_{\downarrow} \times (-\sin x)$

• Algemene vorm van kettingregel

$$\frac{d}{dx} [f(x)]^n = n [f(x)]^{n-1} \times f'(x) *$$

* Maar $f(x) = y$

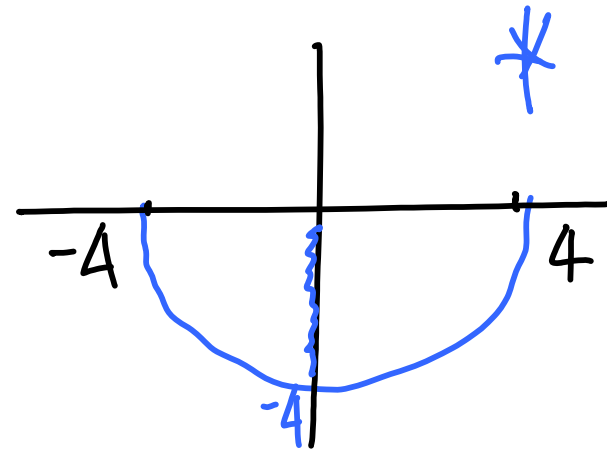
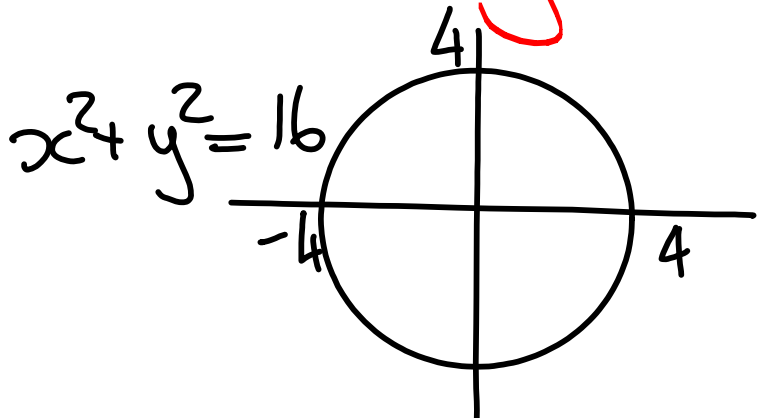
Nou $\frac{d}{dx}(y^n) = n y^{n-1} \times \frac{dy}{dx}$ * $\left(\frac{dy}{dx} = y'\right)$

bv. $\frac{d}{dx}(y^2) = 2y^{2-1} \times \frac{dy}{dx}$

Verskil tussen eksplisiete
en implisiete differensiasie

• Voorbeeld: Bepaal $\frac{dy}{dx}$ as

$x^2 + y^2 = 16$, $y < 0$.



Ekspliciete differentiatie:

• [as $y = \dots$ bepaal $\frac{dy}{dx}$] Δ

• $x^2 + y^2 = 16$ en $y < 0$ (onthou)

$$\Rightarrow y^2 = 16 - x^2$$

$$\Rightarrow y = -\sqrt{16 - x^2}$$

Dan $\frac{dy}{dx} = y' = -\frac{1}{2} (16 - x^2)^{-\frac{1}{2}} \times -2x$

$$\Rightarrow \frac{dy}{dx} = \frac{x}{\sqrt{16 - x^2}} \quad (*)$$

Implizite differenziasie:

Differensieer beide kante !!

(Verband tussen x en y)

$$x^2 + y^2 = 16 \quad \text{en} \quad y < 0 \quad \therefore y = -\sqrt{16 - x^2}$$

$$\frac{d}{dx}(x^2 + y^2) = \frac{d}{dx}(16)$$

$$\Rightarrow 2x + 2y \times \boxed{\frac{dy}{dx}} = 0$$

$$\Rightarrow 2y \frac{dy}{dx} = -2x \Rightarrow \frac{dy}{dx} = \frac{-2x}{2y} = \frac{-x}{y}$$

$$\text{Dan is } \frac{dy}{dx} = \frac{-x}{-\sqrt{16-x^2}} = \frac{x}{\sqrt{16-x^2}} \quad (\text{antw. dieselfde})$$

VOORBEELDE :

Gebruik implisiete differentia-
sie om $\frac{dy}{dx}$ te bepaal. (Plan: *diff beide kante*)

①. $x^2 y^3 = e^y \Rightarrow \frac{d}{dx} (x^2 y^3) = \frac{d}{dx} (e^y)$
(*product rule*)

$$\Rightarrow \boxed{2x \times y^3} + x^2 \times 3y^2 \times \frac{dy}{dx} = e^y \times \frac{dy}{dx}$$
$$\Rightarrow (3x^2 y^2 - e^y) \frac{dy}{dx} = -2xy^3$$
$$\Rightarrow \frac{dy}{dx} = \frac{-2xy^3}{3x^2 y^2 - e^y}$$

$$\textcircled{2} \quad \frac{d}{dx}(y^2 + \sin(2x)) = \frac{d}{dx}(\cos y)$$

$$\Rightarrow 2y \frac{dy}{dx} + \underline{2 \cos(2x)} = -\sin y \times \frac{dy}{dx} \quad \checkmark$$

$$\Rightarrow (2y + \sin y) \frac{dy}{dx} = -2 \cos 2x$$

$$\Rightarrow \frac{dy}{dx} = \frac{-2 \cos 2x}{2y + \sin y} \quad \checkmark$$

$$\textcircled{3} \frac{d}{dx}(x^3 + y^4 + \underbrace{x^5 y^7}_{\text{produkt}}) = \frac{d}{dx}(8)$$

$$\Rightarrow 3x^2 + 4y^3 \frac{dy}{dx} + 5x^4 y^7 + 7x^5 y^6 \frac{dy}{dx} = 0$$

$$\Rightarrow (4y^3 + 7x^5 y^6) \frac{dy}{dx} = -3x^2 - 5x^4 y^7$$

$$\Rightarrow \frac{dy}{dx} = \frac{-3x^2 - 5x^4 y^7}{4y^3 + 7x^5 y^6}$$

Afgeleides van inverse Trigonometrische functies

Stewart bl 213

KEN en MOET KAN BEWYS.

$$\begin{aligned} \textcircled{1} \frac{d}{dx} (\text{bg} \sin x) &= \frac{1}{\sqrt{1-x^2}} \\ \textcircled{2} \frac{d}{dx} (\text{bg} \tan x) &= \frac{1}{1+x^2} \end{aligned} \left. \vphantom{\begin{aligned} \textcircled{1} \frac{d}{dx} (\text{bg} \sin x) &= \frac{1}{\sqrt{1-x^2}} \\ \textcircled{2} \frac{d}{dx} (\text{bg} \tan x) &= \frac{1}{1+x^2} \end{aligned}} \right\} \begin{array}{l} \text{Hoet nie} \\ \text{afgeleides van} \\ \text{ander inverse} \\ \text{trig. functies} \\ \text{te ken nie.} \end{array}$$

Onthou:

$$\begin{aligned} \text{bg} \tan x &= \tan^{-1} x = \arctan x \\ \text{bg} \sin x &= \sin^{-1} x = \arcsin x \end{aligned}$$

① Te bewijs : $\frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}$

Bewijs:

Def. $\arcsin x = y \Leftrightarrow \sin y = x, y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$

Plan: impliciet diff

$$\frac{d}{dx}(\sin y) = \frac{d}{dx}(x)$$

$$\Rightarrow \cos y \times \frac{dy}{dx} = 1$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\cos y}$$

Antwoord

$$\frac{1}{\cos y}$$

• Identität: $\sin^2 y + \cos^2 y = 1$ ↓

Dann ist $\cos^2 y = 1 - \sin^2 y$ ($\cos y \geq 0$)

$\therefore \cos y = \sqrt{1 - \sin^2 y}$ (mit def ist $\sin y = x$)

$= \sqrt{1 - x^2}$

Dann ist $\frac{d}{dx} (\log \sin x) = \frac{1}{\cos y} = \frac{1}{\sqrt{1 - x^2}}$

Te Bewys: • $\frac{d}{dx}(\operatorname{arctan} x) = \frac{1}{1+x^2}$
(Probeer zelf)

Bewys:

Definisie: $y = \operatorname{arctan} x$
 $\Leftrightarrow \tan y = x, \quad y \in (-\frac{\pi}{2}, \frac{\pi}{2})$
(Diff. impliciet)

$$\frac{d}{dx}(\tan y) = \frac{d}{dx}(x)$$

$$\Rightarrow \sec^2 x \frac{dy}{dx} = 1$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\sec^2 x}$$

Identiteit: $\sec^2 y = 1 + \tan^2 y$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{1 + \tan^2 y} \quad (\text{voor } \tan y = x)$$

$$= \frac{1}{1 + x^2}$$

$$\text{Dan is: } \frac{d}{dx}(\log \tan x) = \frac{1}{1 + x^2}$$

Voorbeelde: Bepaal die afgeleides
van die volgende funkties.
Vereenvoudig waar moontlik.

$$f(x) = \sin x \Rightarrow f'(x) = \cos x$$

$$\textcircled{1} f(x) = \sin(x^2)$$

Dan ~~weglaat~~

$$f'(x) = \cos(x^2) \times 2x$$

Geen Punt $\times$

$$f(x) = \log \sin x \Rightarrow f'(x) = \frac{1}{\sqrt{1-x^2}}$$

$$f(x) = \log \sin(x^2)$$

$$f'(x) = \frac{1}{\sqrt{1-(x^2)^2}} \times 2x$$

$$= \frac{2x}{\sqrt{1-x^4}}$$

$$\textcircled{2} f(x) = \tan e^{3x}$$

$$f'(x) = \sec^2(e^{3x}) \times e^{3x} \times 3$$

$$f(x) = \log \tan e^{3x}$$

$$f'(x) = \frac{1}{1+(e^{3x})^2} \times e^{3x} \times 3$$

$$= \frac{3e^{3x}}{1+e^{6x}}$$

③ $y = e^{\text{bg tan}(2x)}$

$y' = e^{\text{bg tan}(2x)} \times \frac{1}{1 + \underbrace{(2x)^2}} \times 2$

④ $y = x \sin^{-1} x + \sqrt{1-x^2}$

produk

Notasie:
 $\sin^{-1} = \text{arcsin}$

$y' = \left(\sin^{-1} x + x \times \frac{1}{\sqrt{1-x^2}} \right) + \frac{1}{\cancel{2}\sqrt{1-x^2}} \times -\cancel{2}x$

⑤ $\tan^{-1}(x^2 y) = x + x y^2$ (Bepaal $\frac{dy}{dx}$ itu x en y)
(implisiet)

Notasie: $\tan^{-1}x = \arctan x$

(lit prakties)

P215 #17

$$\frac{d}{dx} [\tan^{-1}(x^2 y)] = \frac{d}{dx} [x + x y^2]$$

$$\Rightarrow \frac{1}{1+(x^2 y)^2} \times [2xy + x^2 \frac{dy}{dx}] = 1 + [1xy^2 + (x)(2y)\frac{dy}{dx}]$$

$$\Rightarrow \frac{dy}{dx} \left[\frac{x^2}{1+(x^2 y)^2} - 2xy \right] = \frac{2xy}{1+(x^2 y)^2} + 1 + y^2$$

$$\Rightarrow \frac{dy}{dx} = \left[\frac{-2xy}{1+(x^2y)^2} + 1 + y^2 \right] \times \frac{1}{\frac{x^2}{1+(x^2y)^2} - 2xy}$$

$\left(\frac{1}{10} \right)$
 $\approx$